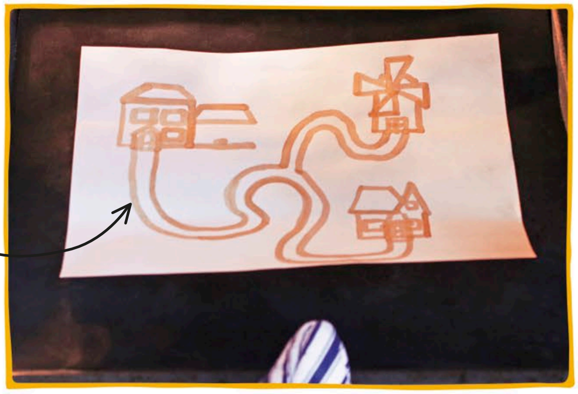
This hidden message was actually a secret map!

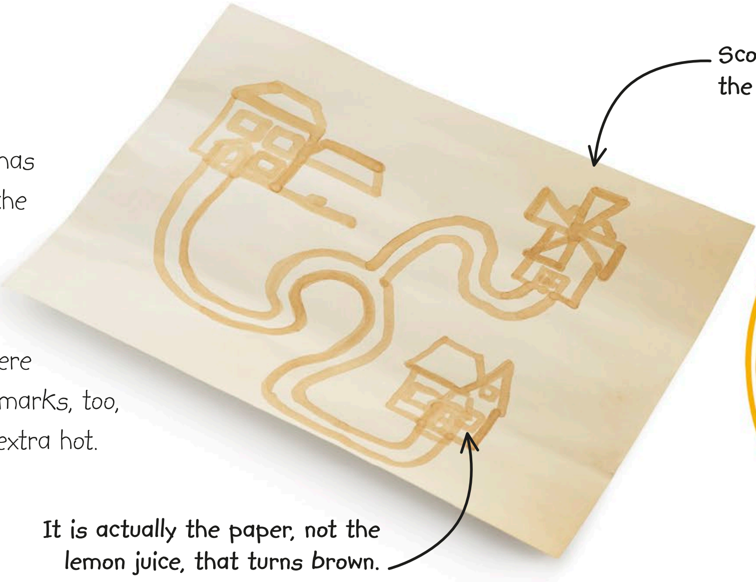


3 With the help of an adult, set the oven to 400°F (200°C). Place your paper on a baking sheet and, when the oven is hot enough, use your oven mitt to put it inside the oven.

4 After half an hour, your invisible markings should have become visible. With an adult's help, use the oven mitt to remove the sheet from the oven and put it on a heatproof surface to cool down.

Scorch marks make the paper look old.

5 Once the sheet has cooled, pick up the paper—it will feel brittle. The heat of the oven dried the paper out, and there may be some scorch marks, too, where the paper got extra hot.



It is actually the paper, not the lemon juice, that turns brown.

Paper naturally goes brown with age—lemon juice and heat speed up the process.

HOW IT WORKS

Paper is made of a compound called cellulose. Each large molecule of cellulose is made up of thousands of smaller molecules of glucose (a type of sugar) that are bonded together. The citric acid in lemon juice slowly weakens the bonds between the glucose molecules, freeing some of them. When the paper is heated to over 350°F (170°C), these free glucose molecules react together in a chemical process called caramelization. This produces new compounds that have a brown color and makes the ink visible.